RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Friday, 25 September 2026, 2:58 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:59 PM
Time taken	40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int hasPairWithDifference(int A[], int n, int k) {
    int i = 0;
    int j = 1;

    while (i < n && j < n) {
        // Indices must be distinct
        if (i == j) {
            j++;
            continue;
        }

        int diff = A[j] - A[i];

        if (diff == k) {
            return 1;
        } else if (diff < k) {
            j++; // Increase the difference by moving the right pointer
        } else {
            i++; // Decrease the difference by moving the left pointer
        }
    }

    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *A = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &A[i]) != 1) {
            free(A);
            return 0;
        }
    }
```

```
    int k;
    if (scanf("%d", &k) != 1) {
        free(A);
        return 0;
    }

    // Find and print result
    printf("%d\n", hasPairWithDifference(A, n, k));

    free(A);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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